

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER

- 1 (a) Work out
- $6 + 4 \div 2$
- .

8 [1]

- (b) Work out
- 40×0.3
- .

12 [1]

- 2 Write these numbers in order of size, starting with the smallest.

$\frac{1}{5}$

$\frac{3}{25}$

13%

0.1

0.1, $\frac{3}{25}$, 13%, $\frac{1}{5}$ [2]
smallest

- 3 (a) Work out the temperature that is 20 degrees higher than
- -12°C
- .

8 $^{\circ}\text{C}$ [1]

- (b) Work out the difference between
- -4°C
- and
- 10°C
- .

14 $^{\circ}\text{C}$ [1]

- 4 Kasia buys 12 apples.
Each apple costs 65 cents.

Work out how much Kasia pays.
Give your answer in dollars.

$$12 \times 65 = 780 \text{ cents}$$

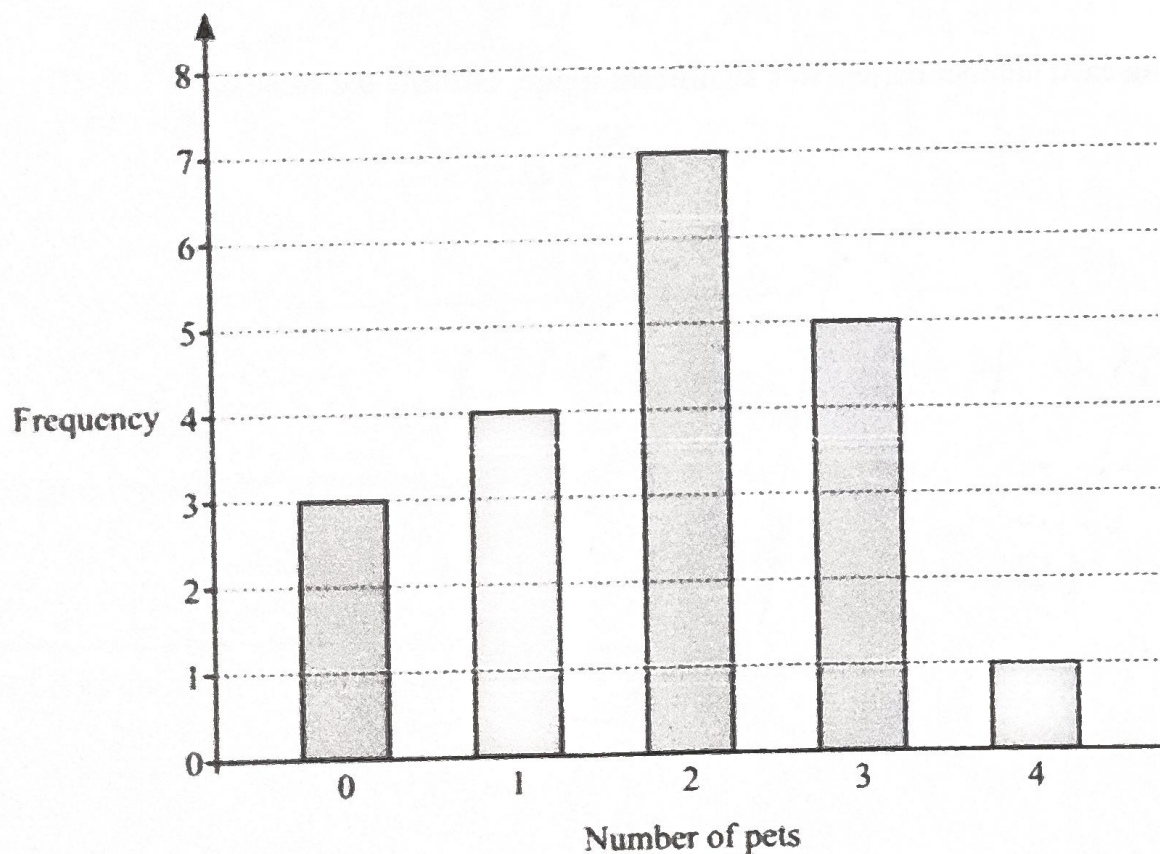
$$\frac{780}{100} = 7.8$$

$$\square \times \square = \square \text{ cents}$$

$$\frac{\square \text{ cents}}{\square} = \$ \underline{\hspace{2cm}}$$

\$ 7.8 [2]

- 5 Yasmin asks 20 people how many pets they own.
The results are shown in the bar chart.



- (a) Find the range.

$$4 - 0 \quad | \quad \square - \square$$

4

[1]

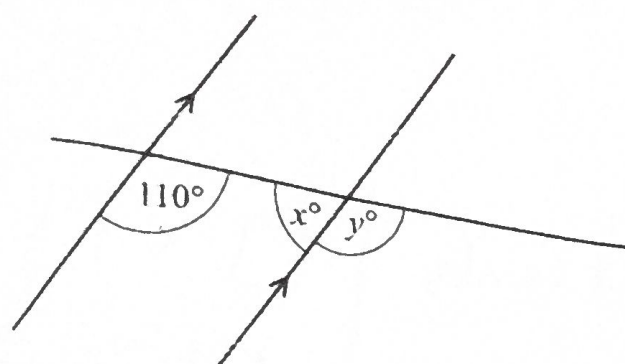
- (b) Find the fraction of the 20 people who own 3 pets.

$$\Rightarrow \frac{5}{20} \quad | \quad \Rightarrow \frac{\square}{\square}$$

$\frac{5}{20}$ or $\frac{1}{4}$

[1]

[Turn over]



NOT TO
SCALE

The diagram shows a straight line crossing two parallel lines.

(a) Work out the value of x .

$$x^\circ = 180 - 110 \quad \left| \begin{array}{l} x = \square - \square \\ x = \square \end{array} \right.$$

$$x = \underline{70} \quad [1]$$

(b) Work out the value of y .

$$y^\circ = 180^\circ - 70^\circ \quad \left| \begin{array}{l} y = \square - \square \\ y = \square \end{array} \right.$$

$$y = \underline{110} \quad [1]$$

7 By writing each number correct to 1 significant figure, estimate the value of

$$\frac{53.7}{2.61 + 7.48}$$

$$\frac{50}{3 + 7}$$

$$\frac{50}{10} = 5$$

$$\frac{\square}{\square + \square}$$

$$\frac{\square}{\square}$$

$$\square$$

$$\underline{5} \quad [2]$$

- 8 (a) Convert 78 mm to cm.

$$\frac{78}{10} = \frac{\boxed{}}{\boxed{}}$$

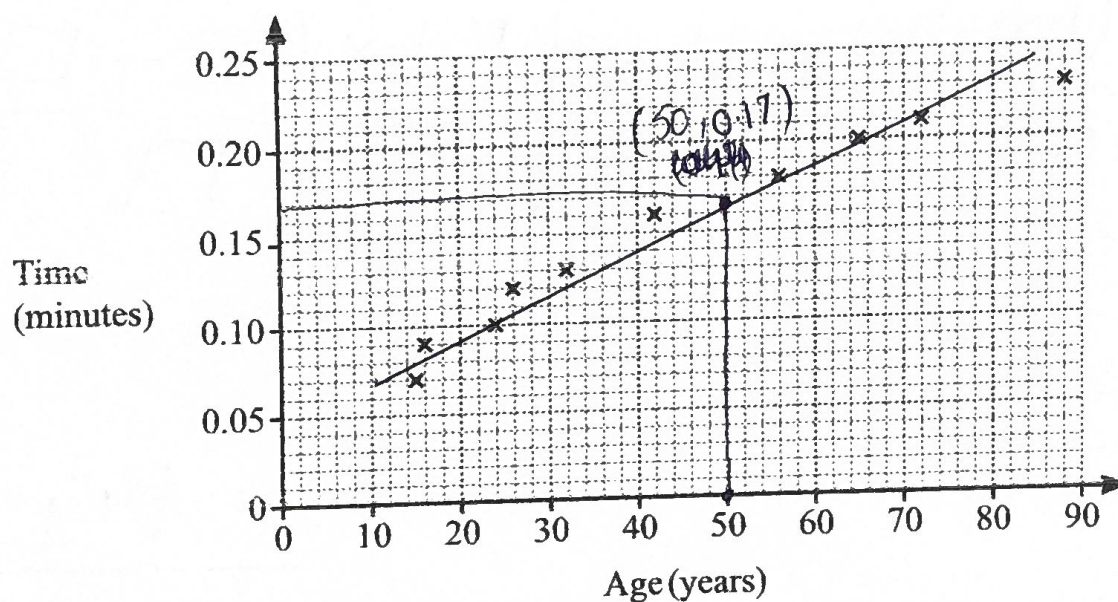
7.8 cm [1]

- (b) Convert 3 m^2 to cm^2 .

$$3 \times (100)^2 = 30000 \quad | \quad \boxed{3} \times (\boxed{})^2 = \boxed{}$$

30000 cm^2 [1]

- 9 The scatter diagram shows the ages of ten people and the time they each take to complete a task.



- (a) Write down the type of correlation shown on the scatter diagram.

Positive [1]

- (b) By drawing a line of best fit, estimate the time taken by a person aged 50 to complete the task.

0.17 minutes [2]

[Turn over]

- 10 (a) Four exterior angles of a pentagon are 150° , 100° , 45° and 35° .

Calculate the size of the remaining exterior angle.

$$\Rightarrow 360 - (150 + 100 + 45 + 35) \quad \left| \quad \square - (\square + \square + \square + \square) \right.$$

$$\Rightarrow 30 \quad \left| \quad \Rightarrow \square \right.$$

30 [2]

- (b) Calculate the interior angle of a regular decagon.

$$\frac{180 \times (10 - 2)}{10} = 144 \quad \left| \quad \frac{\square \times (\square - \square)}{\square} = \square \right.$$

144 [2]

- 11 (a) Evaluate $4^2 + \sqrt[3]{27}$.

$$\Rightarrow 16 + 3 \quad \left| \quad \Rightarrow \square + \square \right.$$

$$\Rightarrow 19 \quad \left| \quad \Rightarrow \square \right.$$

19 [1]

- (b) Evaluate $5^{-1} \times 5^3$.

$$\Rightarrow \frac{1}{5} \times 5^3 \quad \left| \quad \Rightarrow \frac{\square}{\square} \times \square^{\square} \right.$$

$$\Rightarrow 5^2 = 25 \quad \left| \quad \Rightarrow \square \right.$$

25 [2]

13 Work out $1\frac{3}{5} \div 1\frac{2}{3}$.

$$\frac{8}{5} \div \frac{5}{3}$$

$$\frac{8}{5} \times \frac{3}{5}$$

$$\frac{24}{25}$$

$$\frac{\square}{\square} \div \frac{\square}{\square}$$

$$\frac{\square}{\square} \times \frac{\square}{\square}$$

$$24/25$$

[2]

14 (a) Write 36 as a product of its prime factors.

$$\begin{array}{r|l} 2 & 36 \\ \hline 2 & 18 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline & \square \end{array}$$

$$2 \times 2 \times 3 \times 3$$

[2]

(b) Bus A leaves the bus station every 36 minutes.
Bus B leaves the bus station every 48 minutes.
The two buses both leave the bus station at 09 30.

Find the next time when the two buses leave the bus station together.

$$36 = 2 \times 2 \times 3 \times 3$$

$$48 = 2 \times 2 \times 3 \times 2 \times 2$$

$$\text{L.C.M.} = 2 \times 2 \times 3 \times 3 \times 2 \times 2$$

$$\text{L.C.M.} = 144 \text{ min.}$$

$$120 + 24$$

$$2\text{h} + 24\text{min}$$

$$9\text{h} \quad 30\text{min}$$

$$\begin{array}{r} 11 \quad 54 \end{array}$$

$$36 = \square \times \square \times \square \times \square$$

$$48 = \square \times \square \times \square \times \square \times \square$$

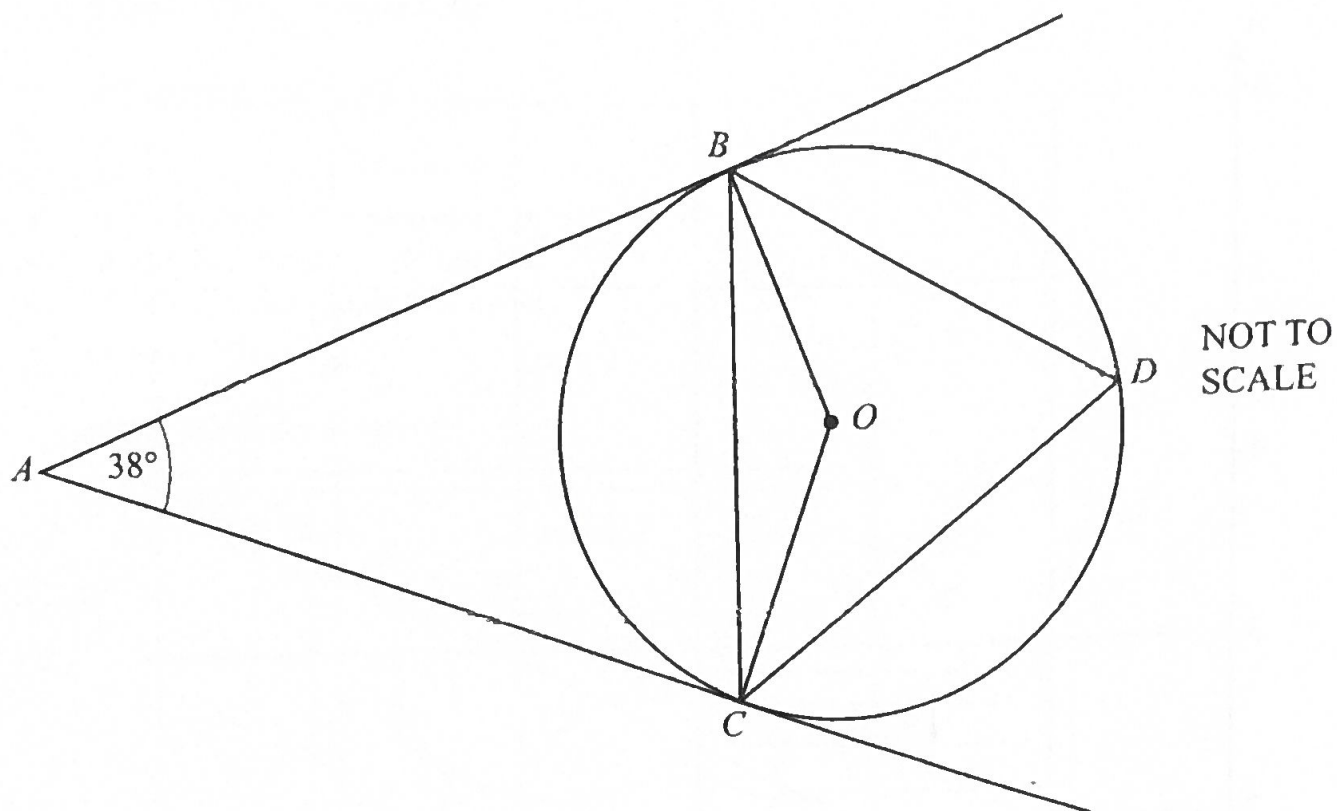
$$\text{L.C.M.} = \square \text{ minutes}$$

$$\text{L.C.M.} = \square \text{h} + \square \text{min}$$

$$\text{Add} = 109 \text{h} + 30 \text{min}$$

$$11 \quad 54$$

[3]



B , C and D are points on the circle, centre O .
 AB and AC are tangents to the circle.
 Angle $BAC = 38^\circ$.

Work out

(a) angle ABC

$$\begin{aligned} \angle A + \angle B + \angle C &= 180^\circ \\ 38 + x + x &= 180 \\ 2x &= 180 - 38 \\ x &= 71 \end{aligned}$$

$$\cancel{138} \quad \square + \square + \square = 180^\circ$$

$$\square = \square$$

Angle $ABC = \dots\dots\dots 71 \dots\dots\dots [1]$

(b) angle BOC

$$\begin{aligned} \Rightarrow 180 - 90 - 19 &= 71 \\ \Rightarrow 71 + 71 &= 142 \end{aligned}$$

$$\Rightarrow \boxed{180} - \square - \square = \square$$

$$\Rightarrow \square + \square = \square$$

$\Rightarrow$ Angle $BOC = \dots\dots\dots 142 \dots\dots\dots [2]$

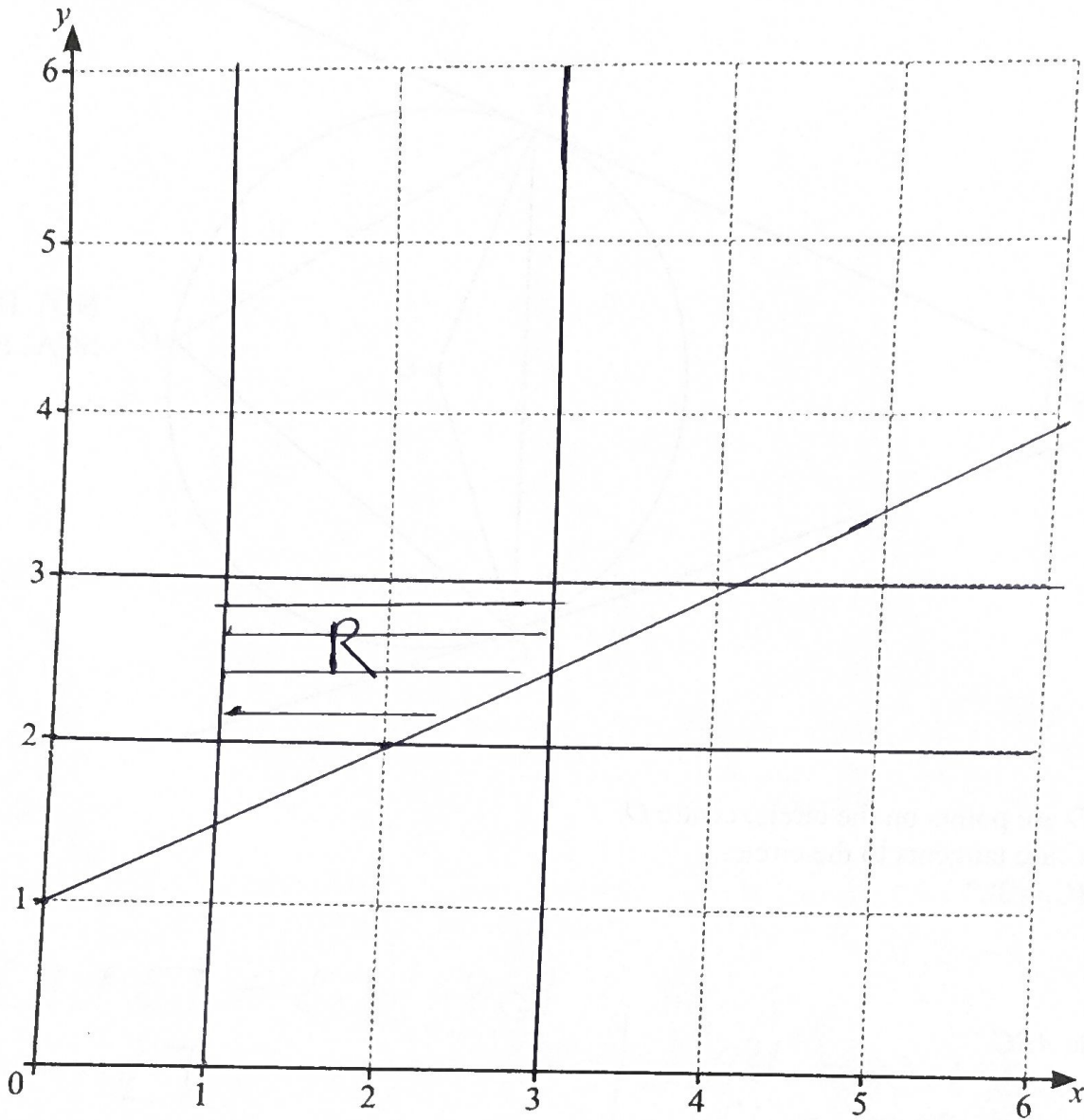
(c) angle BDC .

$$\begin{aligned} \cancel{2 \times 71} &\Rightarrow \frac{142}{2} \\ &\Rightarrow 71 \end{aligned}$$

$$\Rightarrow \frac{\square}{\square} = \square$$

$$\Rightarrow \square$$

$\Rightarrow$ Angle $BDC = \dots\dots\dots 71 \dots\dots\dots [1]$



The region R is defined by these inequalities.

$$1 \leq x \leq 3$$

$$2 \leq y \leq 3$$

$$y \geq \frac{x}{2} + 1$$

Find and label region R .

[4]

- 17 y is directly proportional to the square root of x .
When $x = 16$, $y = 2$.

Find y when $x = 25$.

$$\begin{aligned} y &\propto \sqrt{x} \\ y &= K\sqrt{x} \\ 2 &= K\sqrt{16} \\ 2 &= K \cdot 4 \\ K &= \frac{1}{2} \end{aligned} \quad \left| \quad \begin{aligned} y &= K\sqrt{x} \\ y &= \frac{1}{2}\sqrt{25} \\ y &= \frac{5}{2} \end{aligned} \right.$$

$$\begin{aligned} \square &\propto \square \\ \square &= \square \square \\ \square &= \square \square \\ K &= \frac{\square}{\square} \end{aligned}$$

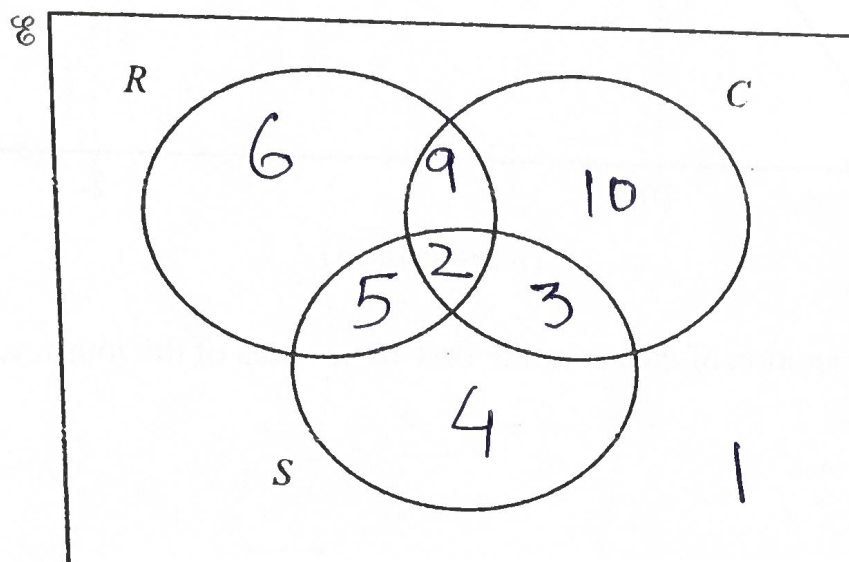
$$\begin{aligned} y &= \square \sqrt{\square} \\ y &= \frac{\square}{\square} \end{aligned}$$

$$y = \dots\dots\dots \frac{5}{2} \quad [2]$$

8 (a) In a sports club of 40 members:

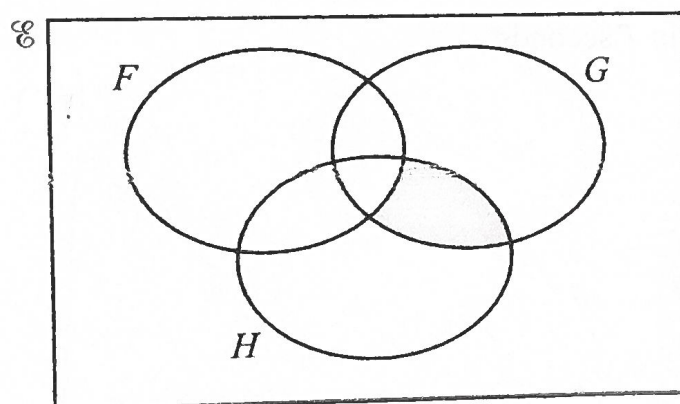
- 22 members run (R)
- 24 cycle (C)
- 14 sail (S)
- 3 cycle and sail but do not run
- 9 run and cycle but do not sail
- 5 run and sail but do not cycle
- 6 run only.

Complete the Venn diagram.



[3]

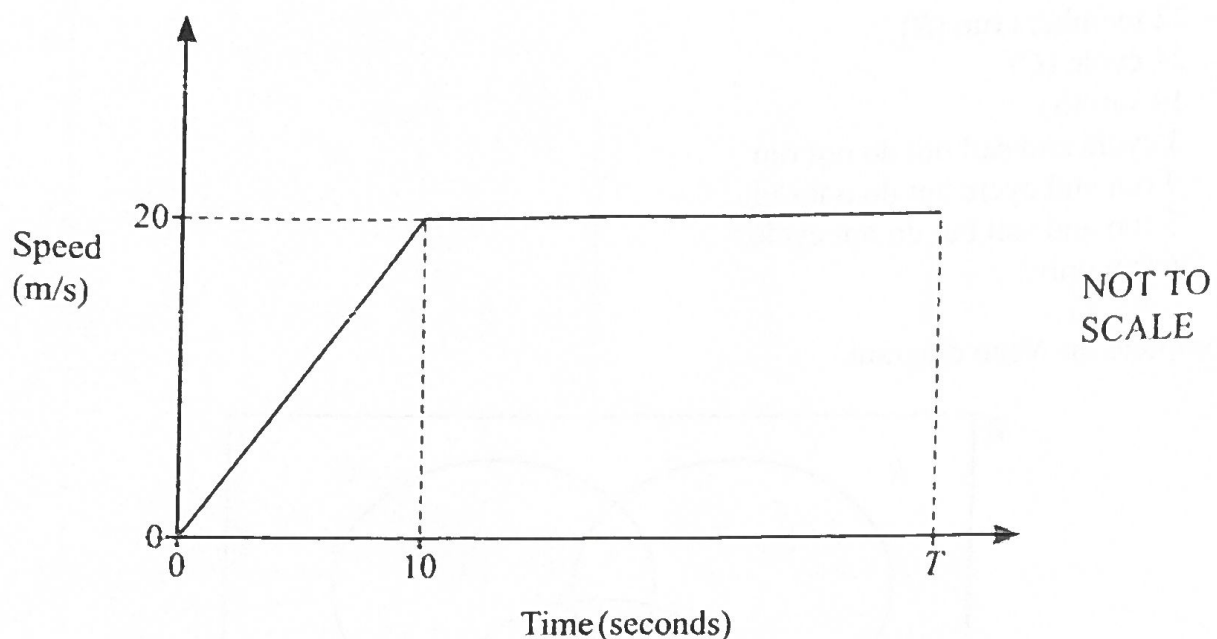
(b) Use set notation to describe the shaded subset in the Venn diagram.



$G \cap H \cap F$ [1]

[Turn over

- 19 The diagram shows the speed-time graph for part of a car journey.



- (a) Calculate the acceleration of the car in the first 10 seconds of the journey.

$$a = \frac{20}{10}$$

$$a = 2$$

$$\Rightarrow a = \frac{\square}{\square}$$

$$a = \square$$

..... 2 m/s² [1]

- (b) The car travels 700 m in T seconds.

Find the value of T .

$$T = \frac{700}{20}$$

$$700 = \frac{1}{2} \times 10 \times 20 + 20(T - 10)$$

$$700 = 100 + 20(T - 10)$$

$$\frac{700 - 100}{20} = T - 10$$

$$30 = T - 10$$

$$T = 40$$

$$\square = \square \times \square \times \square + \square(T - \square)$$

$$T = \square$$

$T =$ 40 [3]

20

$$A = \begin{pmatrix} -2 & 1 \\ 4 & 3 \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}$$

(a) Find A^{-1} .

$$|A| = 3 \times (-2) - (4 \times 1)$$

$$|A| = -6 - 4$$

$$|A| = -10$$

$$\text{Adj} A = \begin{pmatrix} 3 & -1 \\ -4 & -2 \end{pmatrix}$$

$$\begin{array}{l} |A| = \square \times \square - \square \times \square \\ |A| = \square - \square \\ |A| = \square \\ \text{Adj} A = \begin{pmatrix} \square & \square \\ \square & \square \end{pmatrix} \end{array} \quad \left| \quad A^{-1} = \frac{1}{|A|} \begin{pmatrix} \square & \square \\ \square & \square \end{pmatrix} \right.$$

$$-\frac{1}{10} \begin{pmatrix} 3 & -1 \\ -4 & -2 \end{pmatrix} \quad [2]$$

(b) Find AB .

$$\begin{pmatrix} -2 & 1 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -6 + (-1) & -4 + 1 \\ 12 + (-3) & 8 + 3 \end{pmatrix} = \begin{pmatrix} -7 & -3 \\ 9 & 11 \end{pmatrix}$$

$$\begin{array}{l} \begin{pmatrix} \square & \square \\ \square & \square \end{pmatrix} \begin{pmatrix} \square & \square \\ \square & \square \end{pmatrix} = AB \\ \begin{pmatrix} \square + \square & \square + \square \\ \square + \square & \square + \square \end{pmatrix} \\ \begin{pmatrix} \square & \square \\ \square & \square \end{pmatrix} \end{array} \quad \left| \quad \begin{pmatrix} -7 & -3 \\ 9 & 11 \end{pmatrix} \right. \quad [2]$$

21 (a) Factorise $6a - 9$.

$$\underline{3(2a - 3)} \quad [1]$$

(b) Factorise $4b^2 - 25$.

$$\underline{(2b + 5)(2b - 5)} \quad [1]$$

(c) Simplify $\frac{2c^2 - 8c}{2c^2 - 5c - 12}$.

$$\frac{2c(c - 4)}{(2c + 3)(c - 4)}$$

$$\frac{2c}{2c + 3}$$

$$\begin{array}{l} \frac{\square(\square - \square)}{(\square + \square)(\square - \square)} \\ \frac{\square}{\square + \square} \end{array}$$

$$\underline{\frac{2c}{2c + 3}} \quad [3]$$

22 $f(x) = \frac{x}{4} + 3$

$g(x) = 2(x-1)$

(a) Find $f(-8)$.

$$f(-8) = \frac{-8}{4} + 3$$

$$f(-8) = 1$$

$$f(-8) = \frac{\square}{4} + 3$$

$$f(-8) = \square$$

1

..... [1]

(b) Find $f^{-1}(x)$.

$$y = \frac{x}{4} + 3$$

$$x = \frac{y}{4} + 3$$

$$x - 3 = \frac{y}{4}$$

$$4x - 12 = y$$

$$f^{-1}(x) = 4x - 12$$

$$y = \frac{x}{4} + 3$$

$$\square = \square x - \square$$

$$f^{-1}(x) = 4x - 12 \quad [2]$$

(c) $f(p) = g(p+5)$

Find the value of p .

~~P~~

$$f(p) = \frac{p}{4} + 3$$

$$g(p+5) = \cancel{g(2)}$$

$$g(p+5) = 2(p+5-1)$$

$$g(p+5) = 2(p+4)$$

$$\frac{p}{4} + 3 = 2(p+4)$$

$$\frac{p+12}{4} = 2(p+4)$$

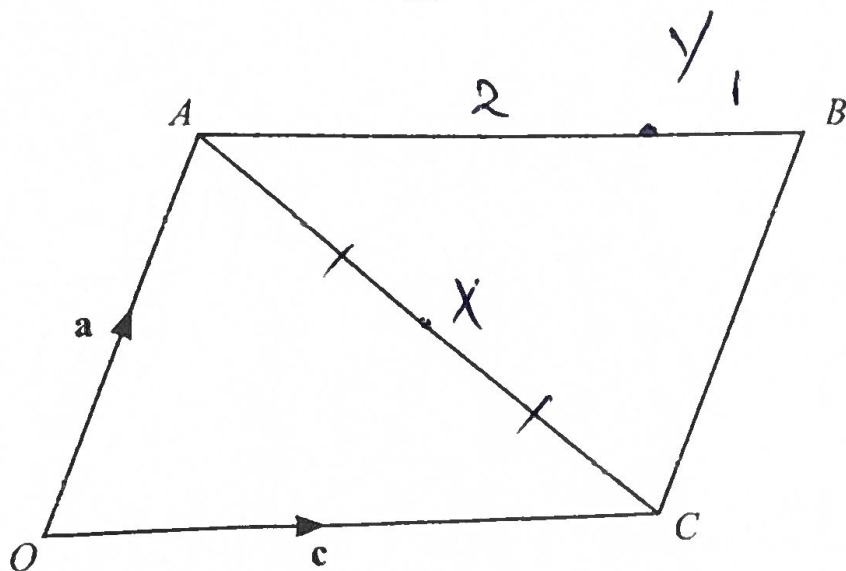
$$p+12 = 8p+32$$

$$p-8p = 32-12$$

$$-7p = 20$$

$$p = -\frac{20}{7}$$

$$p = \cancel{-20/7} \quad [3]$$



NOT TO SCALE

In the diagram, $OABC$ is a parallelogram.
 $\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.

X is the midpoint of AC .
 Y is the point on AB where $AY : YB = 2 : 1$.

Express, as simply as possible, in terms of $\mathbf{a}$ and $\mathbf{c}$

(a) $\vec{AC}$

$$\vec{AC} = \mathbf{c} - \mathbf{a} \quad [1]$$

(b) the position vector of X

$$\vec{OX} = \vec{OC} + \vec{CX}$$

$$\vec{OX} = \mathbf{c} + \left(-\frac{1}{2}\right)(\mathbf{c} - \mathbf{a})$$

$$\vec{OX} = \mathbf{c} - \frac{\mathbf{c}}{2} + \frac{\mathbf{a}}{2}$$

$$\vec{OX} = \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{c}$$

(c) $\vec{YX}$.

$$\vec{YX} = \vec{YA} + \vec{AX}$$

$$\vec{YX} = -\frac{2}{3}\mathbf{c} + \frac{1}{2}(\mathbf{c} - \mathbf{a})$$

$$\vec{YX} = -\frac{2}{3}\mathbf{c} + \frac{1}{2}\mathbf{c} - \frac{1}{2}\mathbf{a}$$

$$\vec{YX} = -\frac{1}{6}\mathbf{c} - \frac{1}{2}\mathbf{a}$$

$$\begin{aligned} \vec{OX} &= \square + \square \\ \vec{OX} &= \square + \square(\square - \square) \\ \vec{OX} &= \square - \square + \square \\ \vec{OX} &= \square + \square \\ &= \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{c} \quad [2] \end{aligned}$$

$$\begin{aligned} \vec{YX} &= \square + \square \\ \vec{YX} &= \square + \frac{\square}{\square}(\square - \square) \\ \vec{YX} &= \square + \square - \square \\ \vec{YX} &= \square - \square \\ \vec{YX} &= -\frac{1}{2}\mathbf{a} - \frac{1}{6}\mathbf{c} \quad [2] \end{aligned}$$

24 Solve $\frac{3x}{x+1} - \frac{2}{x-1} = 3$.

$$\frac{3x(x-1) - 2(x+1)}{(x+1)(x-1)} = 3$$

$$\frac{3x^2 - 3x - 2x - 2}{x^2 - 1} = 3$$

$$3x^2 - 5x - 2 = 3x^2 - 3$$

$$-5x = -3 + 2$$

$$-5x = -1$$

$$x = 1/5$$

$$\frac{x(x-1) - 2(x+1)}{(x+1)(x-1)} = 3$$

$$\frac{x^2 - x - 2x - 2}{x^2 - 1} = 3$$

$$x^2 - 3x - 2 = 3x^2 - 3$$

$$x^2 - 3x - 2 = 3x^2 - 3$$

$$x^2 = 3x^2$$

$$x = \frac{x^2}{x}$$

$$x = \frac{1}{5} \quad [4]$$